PROJECT BASED EXPERIMENTAL LEARNING

COURSE- DATABASE MANAGEMENT SYSTEM (NoSQL / MongoDB)

PROJECT

TOPIC- MONGODB STUDENT DATABASE OPERATIONS

IBM

SUBMITTED TO- MR. Harendra Singh Rajpoot
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1. db.students.insertOne({ rollNo: 101, name: "Rahul", age: 20, department: "CSE", marks: 85 })

```
}
db.thakur.insertOne({ rollNo: 101, name: "Yuvi", age: 20, department: "CSE", marks: 85 })
{
  acknowledged: true,
  insertedId: ObjectId('6abcb2abbd0281f4c346f4e9')
```

2. db.students.insertMany([{ rollNo: 102, name: "Baburao", age: 19, department: "IT", marks: 78 }, { rollNo: 103, name: "Totla seth", age: 21, department: "CSE", marks: 92 }, { rollNo: 104, name: "Modiji", age: 20, department: "ECE", marks: 81 }, { rollNo: 105, name: "Lambu", age: 22, department: "ME", marks: 74 }, { rollNo: 106, name: "Neha", age: 19, department: "CSE", marks: 88 }, { rollNo: 107, name: "Rohit", age: 20, department: "IT", marks: 69 }, { rollNo: 108, name: "Anjali", age: 21, department: "ECE", marks: 95 }, { rollNo: 109, name: "Vikas", age: 20, department: "CSE", marks: 83 }, { rollNo: 110, name: "Pooja", age: 19, department: "IT", marks: 76 }, { rollNo: 111, name: "Karan", age: 22, department: "ME", marks: 72 }, { rollNo: 112, name: "Riya", age: 20, department: "CSE", marks: 89 }, { rollNo: 113, name: "Sahil", age: 21, department: "ECE", marks: 67 }, { rollNo: 114, name: "Kavya", age: 19, department: "IT", marks: 91 }, { rollNo: 115, name: "Nikhil", age: 20, department: "CSE", marks: 79 }, { rollNo: 116, name: "Simran", age: 21, department: "ECE", marks: 84 }, { rollNo: 117, name: "Aditya", age: 22, department: "ME", marks: 73 }, { rollNo: 118, name: "Muskan", age: 20, department: "CSE", marks: 96 }, { rollNo: 119, name: "Varun", age: 19, department: "IT", marks: 68 }, { rollNo: 120, name: "Shreya", age: 21, department: "ECE", marks: 87 }, { rollNo: 121, name: "Manish", age: 20, department: "CSE", marks: 75 }, { rollNo: 122, name: "Nisha", age: 19, department: "IT", marks: 90 }, { rollNo: 123, name: "Akash", age: 22, department: "ME", marks: 71 }, { rollNo: 124, name: "Divya", age: 20, department: "CSE", marks: 86 }, { rollNo: 125, name: "Harsh", age: 21, department: "ECE", marks: 80 }, { rollNo: 126, name: "Isha", age: 19, department: "IT", marks: 94 }, { rollNo: 127, name: "Yash", age: 20, department: "CSE", marks: 77 }, { rollNo: 128, name: "Komal", age: 21, department: "ECE", marks: 82 }, { rollNo: 129, name: "Deepak", age: 22, department: "ME", marks: 65 }, { rollNo: 130, name: "Tanya", age: 20, department: "CSE", marks: 93 }, { rollNo: 131, name: "Mohit", age: 19, department: "IT", marks: 70 }, { rollNo: 132, name: "Sakshi", age: 21, department: "ECE", marks: 89 }, { rollNo: 133, name: "Rajat", age: 20, department: "CSE", marks: 81 }, { rollNo: 134, name: "Payal", age: 19, department: "IT", marks: 85 }, { rollNo: 135, name: "Abhishek", age: 22, department: "ME", marks: 78 }, { rollNo: 136, name: "Aarti", age: 20, department: "CSE", marks: 97 }, { rollNo: 137, name: "Gaurav", age: 21, department: "ECE", marks: 74 }, { rollNo: 138, name: "Preeti", age: 19, department: "IT", marks: 88 }, { rollNo: 139, name: "Suresh", age: 20, department: "CSE", marks: 66 }, { rollNo: 140, name: "Megha", age: 21, department: "ECE", marks: 92 }, { rollNo: 141, name: "Ravi", age: 22, department: "ME", marks: 69 }, { rollNo: 142, name: "Shivani", age: 20, department: "CSE", marks: 84 }, { rollNo: 143, name: "Ankit", age: 19, department: "IT", marks: 79 }, { rollNo: 144, name: "Pallavi", age: 21, department: "ECE", marks: 91 }, { rollNo: 145, name: "Rohan", age: 20, department: "CSE", marks: 87 }, { rollNo: 146, name: "Monika", age: 19, department: "IT", marks: 73 }, { rollNo: 147, name: "Varsha", age: 21, department: "ECE", marks: 86 }, { rollNo: 148, name: "Pankaj", age: 22, department: "ME", marks: 68 }, { rollNo: 149, name: "Ritika", age: 20, department: "CSE", marks: 90 }, { rollNo: 150, name: "Naveen", age: 21, department: "IT", marks: 76 }, { rollNo: 151, name: "Sonia", age: 19, department: "ECE", marks: 83 }])

```
    '22': ObjectId('6abcb2fcbd0281f4c346f500'),
    '23': ObjectId('6abcb2fcbd0281f4c346f501'),
    '24': ObjectId('6abcb2fcbd0281f4c346f502'),
    '25': ObjectId('6abcb2fcbd0281f4c346f503'),
    '26': ObjectId('6abcb2fcbd0281f4c346f504'),
    '27': ObjectId('6abcb2fcbd0281f4c346f505'),
    '28': ObjectId('6abcb2fcbd0281f4c346f506'),
    '29': ObjectId('6abcb2fcbd0281f4c346f507'),
    '30': ObjectId('6abcb2fcbd0281f4c346f508'),
    '31': ObjectId('6abcb2fcbd0281f4c346f509'),
    '32': ObjectId('6abcb2fcbd0281f4c346f50a'),
    '33': ObjectId('6abcb2fcbd0281f4c346f50b'),
    '34': ObjectId('6abcb2fcbd0281f4c346f50c'),
    '35': ObjectId('6abcb2fcbd0281f4c346f50d'),
    '36': ObjectId('6abcb2fcbd0281f4c346f50e'),
    '37': ObjectId('6abcb2fcbd0281f4c346f50f'),
    '38': ObjectId('6abcb2fcbd0281f4c346f510'),
    '39': ObjectId('6abcb2fcbd0281f4c346f511'),
    '40': ObjectId('6abcb2fcbd0281f4c346f512'),
    '41': ObjectId('6abcb2fcbd0281f4c346f513'),
    '42': ObjectId('6abcb2fcbd0281f4c346f514'),
    '43': ObjectId('6abcb2fcbd0281f4c346f515'),
    '44': ObjectId('6abcb2fcbd0281f4c346f516'),
    '45': ObjectId('6abcb2fcbd0281f4c346f517'),
    '46': ObjectId('6abcb2fcbd0281f4c346f518'),
    '47': ObjectId('6abcb2fcbd0281f4c346f519'),
    '48': ObjectId('6abcb2fcbd0281f4c346f51a'),
    '49': ObjectId('6abcb2fcbd0281f4c346f51b')
  }
```

```
db.thakur.insertMany([ { rollNo: 102, name: "Babu rao", age: 19, department: "IT", marks
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abcb2fcbd0281f4c346f4ea'),
    '1': ObjectId('6abcb2fcbd0281f4c346f4eb'),
    '2': ObjectId('6abcb2fcbd0281f4c346f4ec'),
    '3': ObjectId('6abcb2fcbd0281f4c346f4ed'),
    '4': ObjectId('6abcb2fcbd0281f4c346f4ee'),
    '5': ObjectId('6abcb2fcbd0281f4c346f4ef'),
    '6': ObjectId('6abcb2fcbd0281f4c346f4f0'),
    '7': ObjectId('6abcb2fcbd0281f4c346f4f1'),
    '8': ObjectId('6abcb2fcbd0281f4c346f4f2'),
    '9': ObjectId('6abcb2fcbd0281f4c346f4f3'),
    '10': ObjectId('6abcb2fcbd0281f4c346f4f4'),
    '11': ObjectId('6abcb2fcbd0281f4c346f4f5'),
    '12': ObjectId('6abcb2fcbd0281f4c346f4f6'),
    '13': ObjectId('6abcb2fcbd0281f4c346f4f7'),
    '14': ObjectId('6abcb2fcbd0281f4c346f4f8'),
    '15': ObjectId('6abcb2fcbd0281f4c346f4f9'),
    '16': ObjectId('6abcb2fcbd0281f4c346f4fa'),
    '17': ObjectId('6abcb2fcbd0281f4c346f4fb'),
```

3. db.thakur.updateOne({ rollNo: 101 }, { $set: { marks: 90 } })

```
db.thakur.updateOne( { rollNo: 101 }, { $set: { marks: 90 } } )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

4. db.thakur.updateMany({ department: "CSE" }, { $set: { status: "Active" } })

```
db.thakur.updateMany( { department: "CSE" }, { $set: { status: "Active" } } )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 17,
  modifiedCount: 17,
  upsertedCount: 0
}
```

5. db.thakur.deleteOne({ rollNo: 151 })

```
db.thakur.deleteOne({ rollNo: 151 })
{
  acknowledged: true,
  deletedCount: 1
}
```

6. db.thakur.deleteMany({ marks: { $lt: 40 } })

```
db.thakur.deleteMany({ marks: { $lt: 40 } })
{
  acknowledged: true,
  deletedCount: 0
}
```

7. db.thakur.find({ marks: { $gt: 80 } })

```
> db.thakur.find({ marks: { $gt: 80 } })
< {
    _id: ObjectId('6abcb2abbd0281f4c346f4e9'),
    rollNo: 101,
    name: 'Yuvi',
    age: 20,
    department: 'CSE',
    marks: 90,
    status: 'Active'
  }
```

8. db.thakur.find({ marks: { $lt: 80 } })

```
db.thakur.find({ marks: { $lt: 80 } })
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ea'),
  rollNo: 102,
  name: 'Babu rao',
  age: 19,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ed'),
  rollNo: 105,
  name: 'Lambu',
  age: 22,
  department: 'ME',
  marks: 74
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ef'),
  rollNo: 107,
  name: 'Rohit',
  age: 20,
  department: 'IT',
  marks: 69
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4f2'),
```

9. db.thakurfind({ marks { $eq: 85 } })

```
db.thakur.find({ marks: { $eq: 85 } })
{
  _id: ObjectId('6abcb2fcbd0281f4c346f50a'),
  rollNo: 134,
  name: 'Payal',
  age: 19,
  department: 'IT',
  marks: 85
}
```

10. db.thakurfind({ marks { $gte: 80 } })

```
db.thakur.find({ marks: { $gte: 80 } })
{
  _id: ObjectId('6abcb2abbd0281f4c346f4e9'),
  rollNo: 101,
  name: 'Yuvi',
  age: 20,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
```

11. db.thakurfind({ marks { $ne: 85 } })

```
> db.thakur.find({ marks: { $ne: 85 } })
< {
    _id: ObjectId('6abcb2abbd0281f4c346f4e9'),
    rollNo: 101,
    name: 'Yuvi',
    age: 20,
    department: 'CSE',
    marks: 90,
    status: 'Active'
  }
  {
    _id: ObjectId('6abcb2fcbd0281f4c346f4ea'),
    rollNo: 102,
    name: 'Babu rao',
    age: 19,
    department: 'IT',
    marks: 78
  }
  {
    _id: ObjectId('6abcb2fcbd0281f4c346f4eb'),
    rollNo: 103,
    name: 'Laila',
    age: 21,
    department: 'CSE',
    marks: 92,
    status: 'Active'
  }
  {
    _id: ObjectId('6abcb2fcbd0281f4c346f4ec'),
```

12. db.thakurfind({ $and: [{ marks { $gt: 70 } }, { age: { $lt: 23 } }]})

```
db.thakur.find({ $and: [ { marks: { $gt: 70 } }, { age: { $lt: 23 } } ] })
{
  _id: ObjectId('6abcb2abbd0281f4c346f4e9'),
  rollNo: 101,
  name: 'Yuvi',
  age: 20,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ea'),
  rollNo: 102,
  name: 'Babu rao',
  age: 19,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4eb'),
  rollNo: 103,
  name: 'Laila',
  age: 21,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ec')
```

13. db.thakurfind({ $or: [{ department:"CSE" }, { department:"ECE' }] })

```
db.thakur.find({ $or: [ { department: "CSE" }, { department: "ECE" } ] })
{
  _id: ObjectId('6abcb2abbd0281f4c346f4e9'),
  rollNo: 101,
  name: 'Yuvi',
  age: 20,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4eb'),
  rollNo: 103,
  name: 'Laila',
  age: 21,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ec'),
  rollNo: 104,
  name: 'Totla seth',
  age: 20,
  department: 'ECE',
  marks: 81
}
{
```

14. db.thakurfind({ marks { $not: { $gt: 80 } } })

```
db.thakur.find({ marks: { $not: { $gt: 80 } } })
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ea'),
  rollNo: 102,
  name: 'Babu rao',
  age: 19,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ed'),
  rollNo: 105,
  name: 'Lambu',
  age: 22,
  department: 'ME',
  marks: 74
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4ef'),
  rollNo: 107,
  name: 'Rohit',
  age: 20,
  department: 'IT',
  marks: 69
}
{
  _id: ObjectId('6abcb2fcbd0281f4c346f4f2'),
  rollNo: 110,
  name: 'Pooja',
```